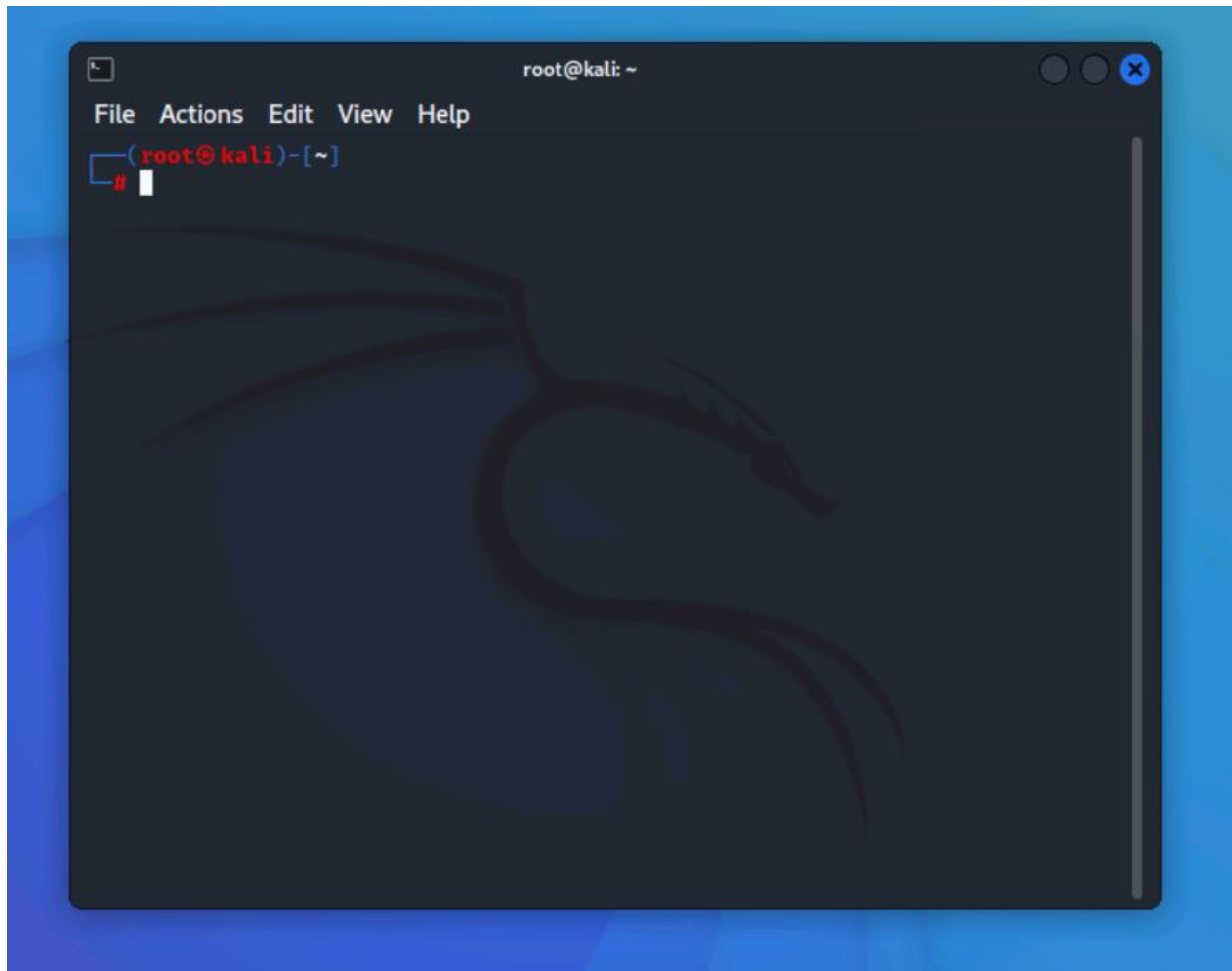
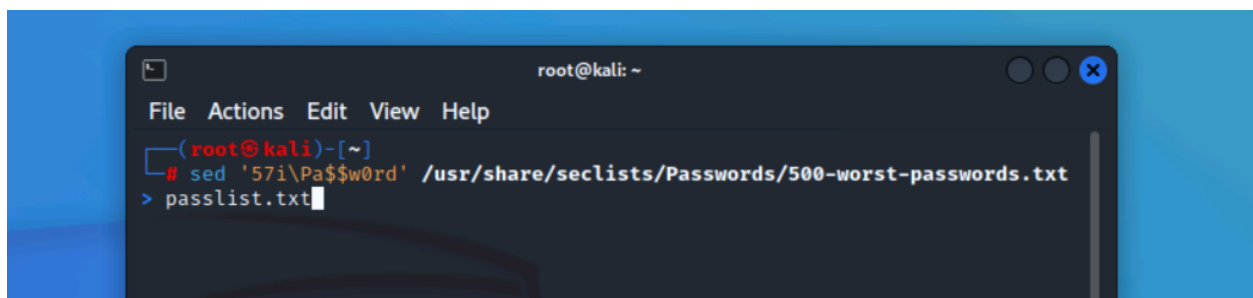


Dominic Amendola

CYB244 IoC Detection & Analysis Lab



I first opened the Kali VM and logged in with the credentials.



Then I typed **sed '57i\Pa\$\$w0rd' /usr/share/seclists/Passwords/500-worst-passwords.txt > passlist.txt** which made a passlist.txt file in the root directory adding Pa\$\$w0rd in the 57th line position of the /usr/share/seclists/Passwords/500-worst-passwords.txt file.

```
> passlist.txt

(root@kali)~# ls -l
total 40
drwxr-xr-x 2 root root 4096 Feb  6 2023 Desktop
drwxr-xr-x 2 root root 4096 Feb  6 2023 Documents
drwxr-xr-x 2 root root 4096 Feb  6 2023 Downloads
-rw-r--r-- 1 root root    0 Feb 19 2023 history
drwxr-xr-x 2 root root 4096 Nov 17 2022 LOD
drwxr-xr-x 2 root root 4096 Feb  6 2023 Music
-rw-r--r-- 1 root root 3500 Mar 25 15:21 passlist.txt
drwxr-xr-x 2 root root 4096 Feb  6 2023 Pictures
drwxr-xr-x 2 root root 4096 Feb  6 2023 Public
drwxr-xr-x 2 root root 4096 Feb  6 2023 Templates
drwxr-xr-x 2 root root 4096 Feb  6 2023 Videos

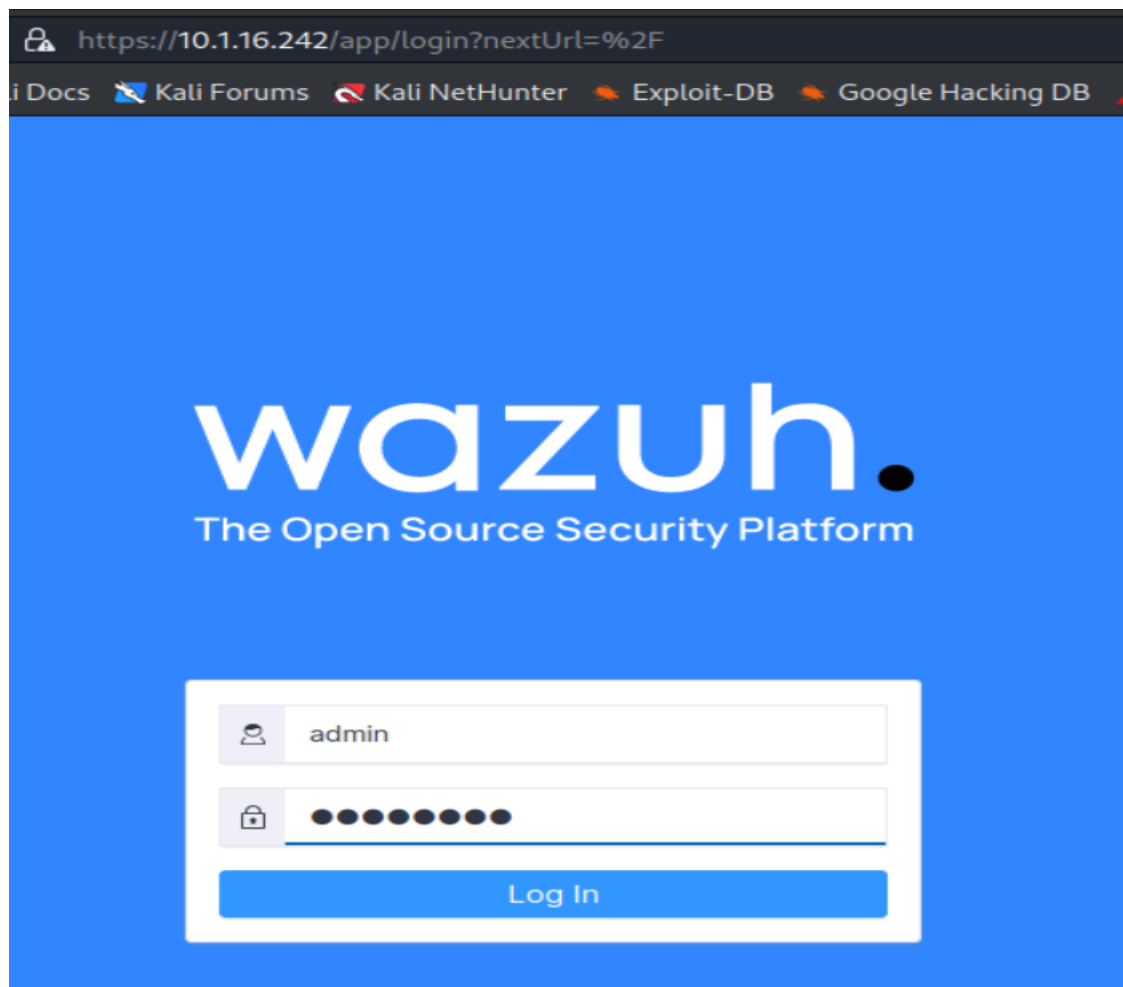
(root@kali)~#
```

Next I typed `ls -l` to confirm the `passlist.txt` file is present.

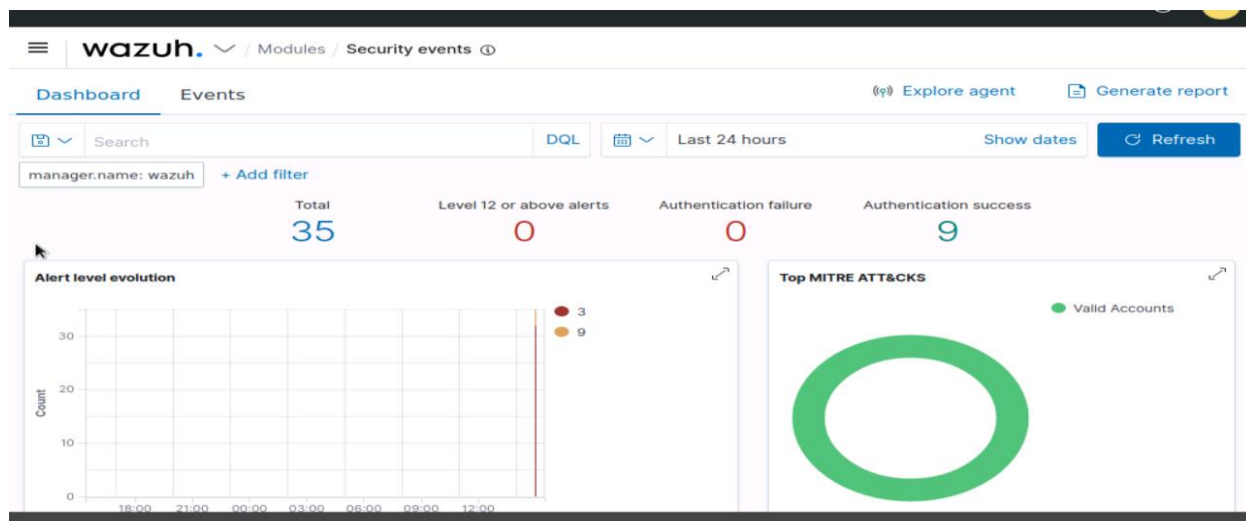
```
(root@kali)~# grep -n 'Pa$$w0rd' passlist.txt
57:Pa$$w0rd

(root@kali)~#
```

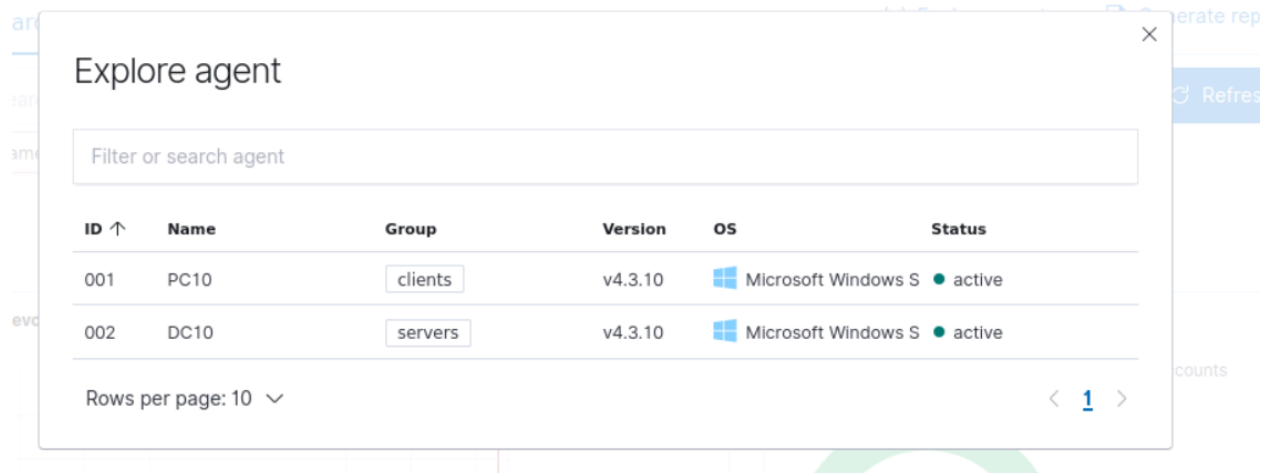
I next typed `grep -n 'Pa$$w0rd' passlist.txt` to confirm the lab password was added to the password list file.



I next opened the Wazuh platform and logged in.



I next viewed the security events on the DC10 VM by clicking the 'security events' tab when Wazuh loaded.



Next, I clicked 'explore agent' at the top of the page and chose DC10 and looked at some of the events.

```
(root@kali)-[~]
# hydra -t 1 -V -f -l administrator -P passlist.txt rdp://10.1.16.1

00 [child 0] (0/0)
[ATTEMPT] target 10.1.16.1 - login "administrator" - pass "Pa$$w0rd" - 57 of
500 [child 0] (0/0)
[3389][rdp] host: 10.1.16.1 login: administrator password: Pa$$w0rd
[STATUS] attack finished for 10.1.16.1 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-03-25 15:
25:57

(root@kali)-[~]
#
```

Back to the terminal, I next typed **hydra -t 1 -V -f -l administrator -P passlist.txt rdp://10.1.16.1** which did a password guessing attempt against the target with 57 attempts, and the 57th one succeeding.



I then went back to wazuh and refreshed the page which then displayed the new events with DC10, and it showed the action I did with hydra in terminal.

```

Ubuntu 20.04.5 LTS wazuh tty1
wazuh login: wazuh
Password:
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-132-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

System information as of Tue 25 Mar 2025 10:29:36 PM UTC

System load:          0.07
Usage of /:            22.2% of 38.34GB
Memory usage:         77%
Swap usage:           0%
Processes:            120
Users logged in:      0
IPv4 address for eth0: 10.1.16.242
IPv6 address for eth0: fdf0:2413:6d1c:20:15:5dff:fe28:f4c4

0 updates can be applied immediately.

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sun Mar 26 22:14:30 UTC 2023 on tty1
wazuh@wazuh:~$ _

```

```

wazuh@wazuh:~$ cd /var/ossec/ruleset/rules
wazuh@wazuh:/var/ossec/ruleset/rules$ _

```

I then switched over to the wazuh VM and logged in, and typed `cd /var/ossec/ruleset/rules` which went into this directory.

```
-rw-r----- 1 root wazuh 1320 Nov 11 2022 0715-freepbx_rules.xml
-rw-r----- 1 root wazuh 119055 Nov 11 2022 0750-github_rules.xml
-rw-r----- 1 root wazuh 52453 Nov 11 2022 0755-office365_rules.xml
-rw-r----- 1 root wazuh 7417 Nov 11 2022 0770-gitlab_rules.xml
-rw-r----- 1 root wazuh 775 Nov 11 2022 0775-arbor_rules.xml
-rw-r----- 1 root wazuh 3980 Nov 11 2022 0780-fireeye_rules.xml
-rw-r----- 1 root wazuh 1744 Nov 11 2022 0785-huawei-usg_rules.xml
-rw-r----- 1 root wazuh 9672 Nov 11 2022 0800-sysmon_id_1.xml
-rw-r----- 1 root wazuh 2413 Nov 11 2022 0810-sysmon_id_3.xml
-rw-r----- 1 root wazuh 1837 Nov 11 2022 0820-sysmon_id_7.xml
-rw-r----- 1 root wazuh 2952 Nov 11 2022 0830-sysmon_id_11.xml
-rw-r----- 1 root wazuh 3322 Nov 11 2022 0840-win_event_channel.xml
-rw-r----- 1 root wazuh 2046 Nov 11 2022 0850-audit_rules.xml
-rw-r----- 1 root wazuh 1233 Nov 11 2022 0860-sysmon_id_13.xml
-rw-r----- 1 root wazuh 472 Nov 11 2022 0870-sysmon_id_8.xml
-rw-r----- 1 root wazuh 1243 Nov 11 2022 0900-firewall_rules.xml
-rw-r----- 1 root wazuh 8431 Nov 11 2022 0905-cisco-ftd_rules.xml
-rw-r----- 1 root wazuh 4407 Nov 11 2022 0910-ms-exchange-proxylogon_rules.xml
-rw-r----- 1 root wazuh 1639 Nov 11 2022 0915-win-powershell_rules.xml
-rw-r----- 1 root wazuh 517 Nov 11 2022 0920-oracledb_rules.xml
-rw-r----- 1 root wazuh 2624 Nov 11 2022 0925-eset-remote_rules.xml
-rw-r----- 1 root wazuh 4026 Nov 11 2022 0935-cloudflare-waf_rules.xml
wazuh@wazuh:/var/ossec/ruleset/rules$
```

```
wazuh@wazuh:/var/ossec/ruleset/rules$ grep 92652 *
0840-win_event_channel.xml: <rule id="92652" level="6">
wazuh@wazuh:/var/ossec/ruleset/rules$ _
```

I next typed `ls -l` to view the rules in this directory and `grep 92652` which displayed the rule contents for 92652.

```

<rule id="92651" level="0">
  <if_sid>60106</if_sid>
  <field name="win.eventdata.ipAddress" type="pcr2">{?:[0-9]{1,3}\.){3}[0-9]{1,3}</field>
  <field name="win.eventdata.ipAddress" type="pcr2" negate="yes">127\.\.\.1</field>
  <description>Successful Remote Logon by user:$(win.eventdata.targetDomainName)\$(win.eventdata.targetUserName)
  <mitre>
    <id>T1078</id>
  </mitre>
  <group>authentication_success,gdpr_IV_32.2,gpg13_7.1,gpg13_7.2,hipaa_164.312.b,nist_800_53_AC.7,nist_800_53_
  .2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
</rule>

<rule id="92652" level="6">
  <if_sid>92651</if_sid>
  <field name="win.eventdata.authenticationPackageName" type="pcr2">NTLM</field>
  <description>Successful Remote Logon Detected - NTLM authentication, possible pass-the-hash attack.</descrip
  <mitre>
    <id>T1550.002</id>
  </mitre>
  <group>authentication_success,gdpr_IV_32.2,gpg13_7.1,gpg13_7.2,hipaa_164.312.b,nist_800_53_AC.7,nist_800_53_
  .2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
</rule>

<rule id="92653" level="3">
0840-win_event_channel.xml

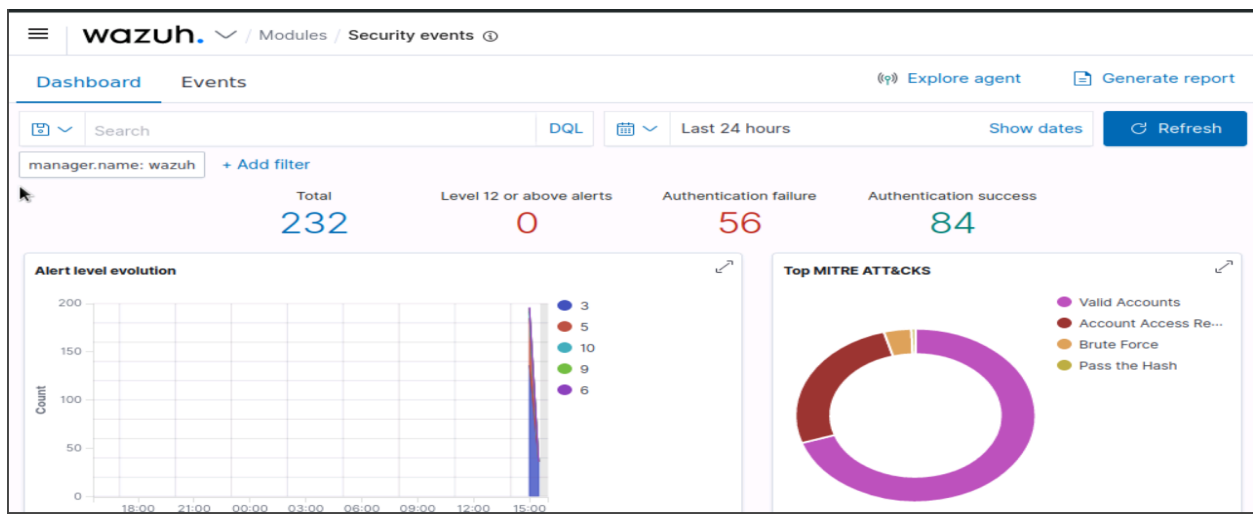
```

```

<rule id="92652" level="6">
  <if_sid>92651</if_sid>
  <field name="win.eventdata.authenticationPackageName" type="pcr2">NTLM</field>
  <description>Successful Remote Logon Detected - NTLM authentication, possible pass-the-hash attack.</description>
  <mitre>
    <id>T1550.002</id>
  </mitre>
  <group>authentication_success,gdpr_IV_32.2,gpg13_7.1,gpg13_7.2,hipaa_164.312.b,nist_800_53_AC.7,nist_800_53_AU.14,pci_dss_10
  .2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,</group>
</rule>

```

Next, I typed **less 0840-win_event_channel.xml** and located the rule with ID 92652 and read it over here.



I then returned the security events in wazuh.

```
(root@kali)-[~]
# mkdir /mnt/dc10-c

(root@kali)-[~]
#
```

```
(root@kali)-[~]
# mount -o username=jaime //10.1.16.1/c$ /mnt/dc10-c
Password for jaime@//10.1.16.1/c$:
mount error(13): Permission denied
Refer to the mount.cifs(8) manual page (e.g. man mount.cifs) and kernel log messages (dmesg)

(root@kali)-[~]
#
```

Back to the terminal, I typed **mount -o username=jaime //10.1.16.1/c\$ /mnt/dc10-c** which attempted the mount the C\$ share using the jaime account and Pa\$\$w0rd as the password, as it should, it failed.

```
(root@kali)-[~]
# mount -o username=administrator //10.1.16.1/c$ /mnt/dc10-c
Password for administrator@//10.1.16.1/c$:

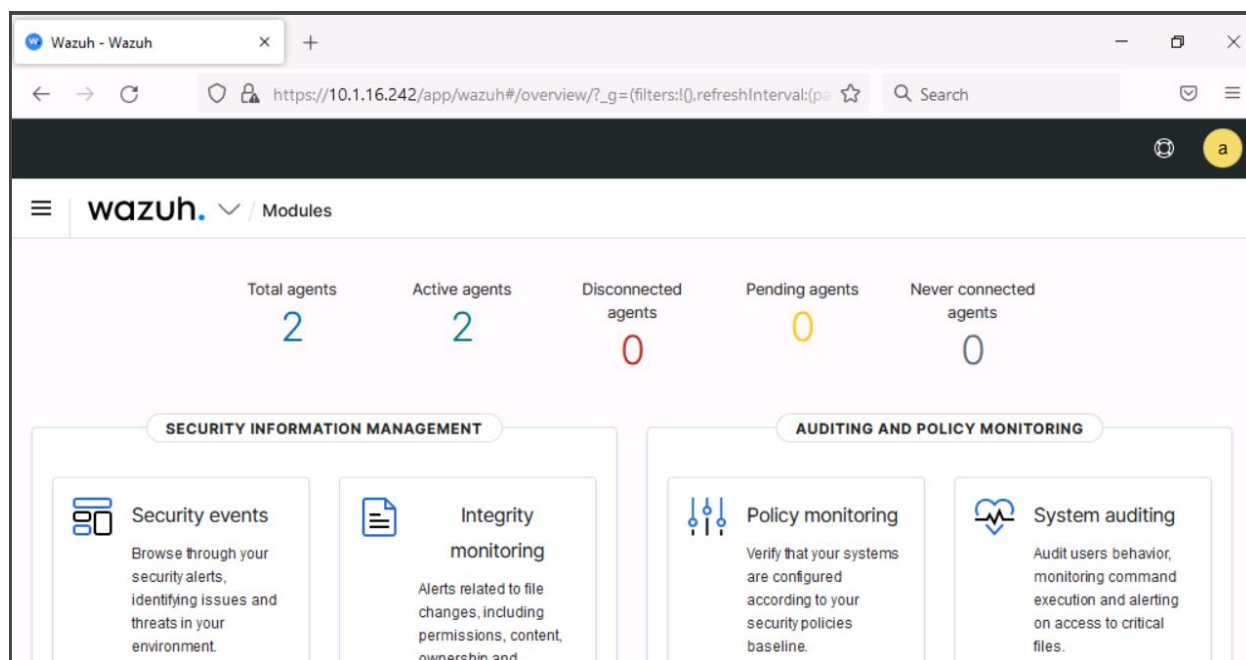
(root@kali)-[~]
#
```

I next typed **mount -o username=administrator //10.1.16.1/c\$ /mnt/dc10-c** which did the same thing, and this mount succeeded.

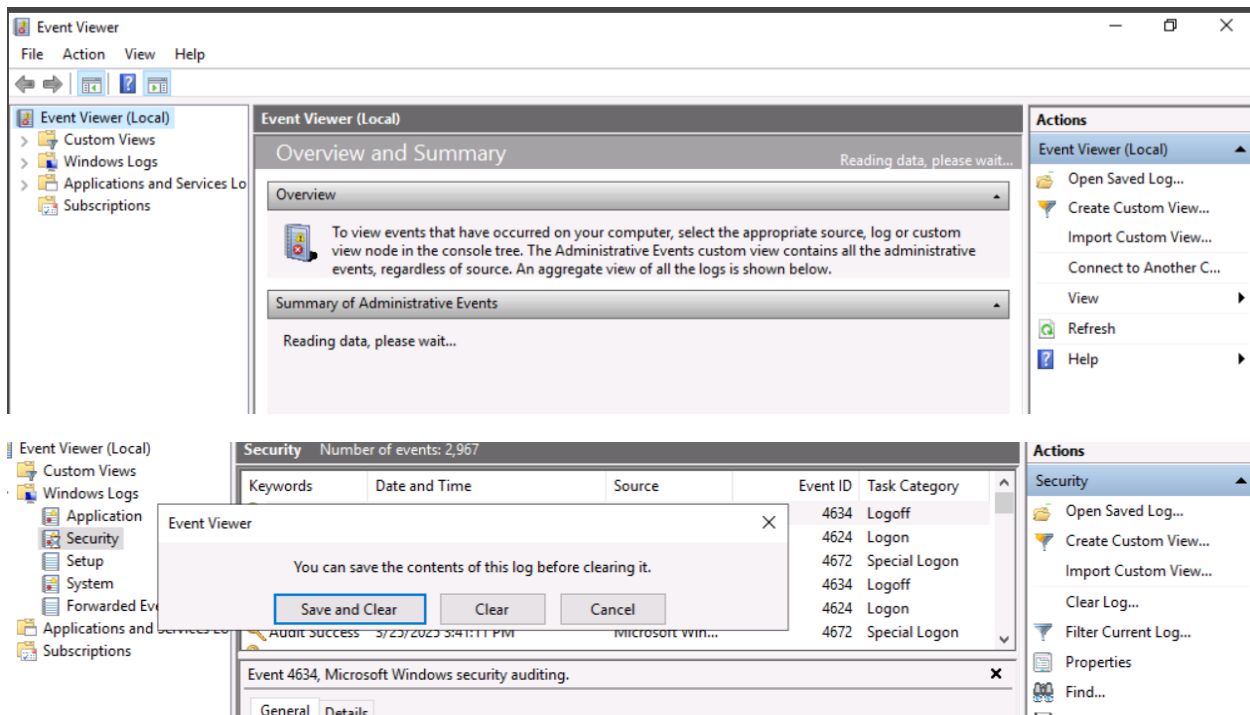
Mar 25, 2025			T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password. 5	60122
> @	002	DC10				
15:25:55.056			T1531			

Mar 25, 2025	002	DC10	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
--------------	-----	------	-------	---	------------------------	---	-------

I then refreshed the wazuh page and then located the 2 rule IDs listed in the lab to locate and I found them, being 60122 and 60106.



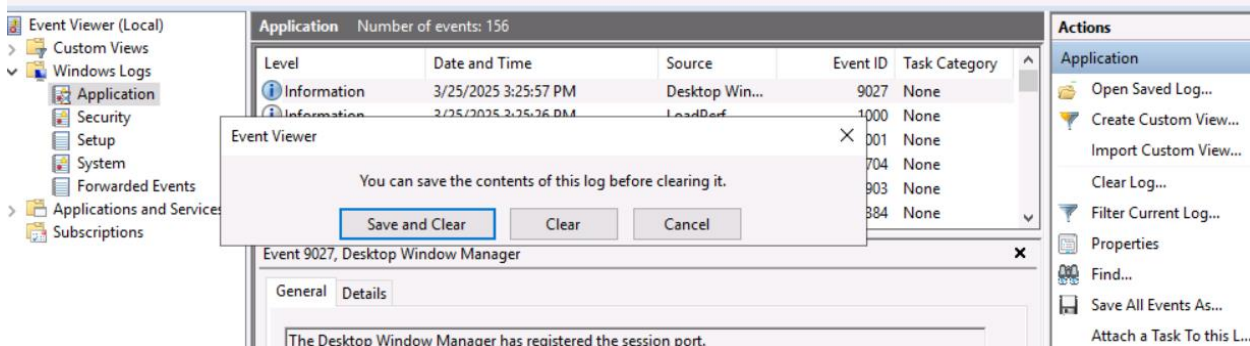
Next, I moved over to the windows VM and opened Wazuh on that VM.



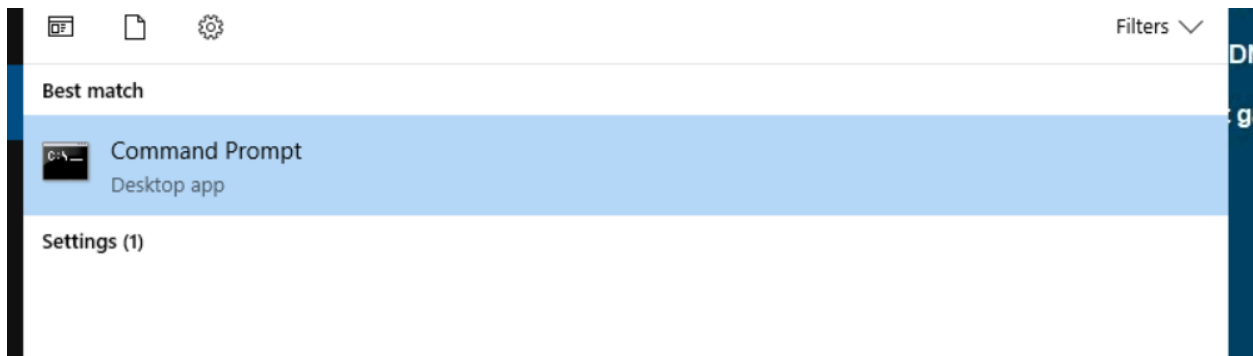
Next on the Windows VM, I opened the event viewer from searching for it in the search bar, and I clicked windows logs > security > clear log > cleared the log.

Mar 25, 2025							
> @ 15:42:56.387	002	DC10	T1070	Defense Evasion	The audit log was cleared	5	63103

After clearing the logs, I went back to wazuh and searched for the ID stating that the audit log was cleared and found it, with its ID being 63103.



Back to event viewer, I went to windows logs > application > clear log > cleared the application logs.



```
C:\Windows\system32>auditpol /set /category:* /success:enable /failure:enable
The command was successfully executed.

C:\Windows\system32>_
```

```
C:\Windows\system32>auditpol /get /category:*
System audit policy
Category/Subcategory          Setting
System
  Security System Extension    Success and Failure
  System Integrity             Success and Failure
  IPsec Driver                 Success and Failure
  Other System Events          Success and Failure
  Security State Change        Success and Failure
Logon/Logoff
  Logon                       Success and Failure
  Logoff                      Success and Failure
  Account Lockout              Success and Failure
  IPsec Main Mode              Success and Failure
  IPsec Quick Mode             Success and Failure
  IPsec Extended Mode          Success and Failure
  Special Logon                Success and Failure
  Other Logon/Logoff Events     Success and Failure
  Network Policy Server         Success and Failure
  User / Device Claims          Success and Failure
  Group Membership              Success and Failure
Object Access
  File System                  Success and Failure
  Registry                     Success and Failure
```

To the DC10 VM, I opened CMD as admin and typed **auditpol /set /category:* /success:enable /failure:enable** and **auditpol /get /category:*** to turn on the audit policies.

```
C:\Windows\system32>net users

User accounts for \\DC10

-----
Administrator      Angel              Ca
Cam                 Dani              Dylan
Guest              Jaime             Jaylan
Kei                 krbtgt            Rene
Sam                 Viral
The command completed successfully.

C:\Windows\system32>
```

I next typed **net users** to view the users on this VM.

```
C:\Windows\system32>net users /delete Angel
The command completed successfully.

C:\Windows\system32>
```

```
C:\Windows\system32>net users /delete Dylan
The command completed successfully.

C:\Windows\system32>
```

Next, I typed **net users /delete <name>** to delete the users angel and dylan.

```
C:\Windows\system32>net users

User accounts for \\DC10

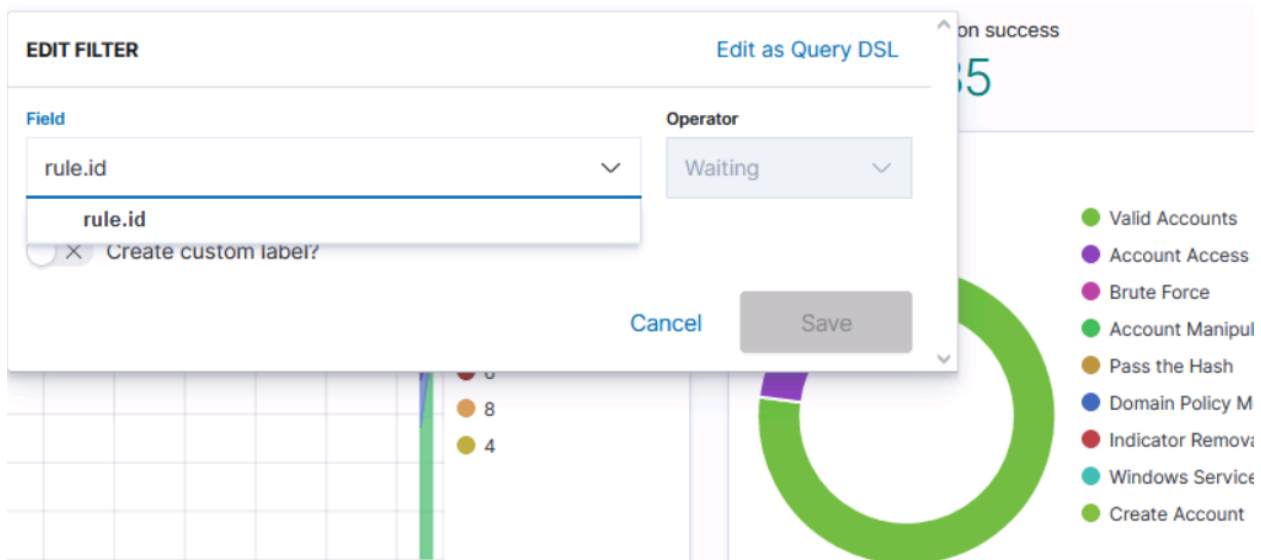
-----
Administrator      ca                Cam
Dani                Guest             Jaime
Jaylan              Kei               krbtgt
Rene                Sam               Viral
The command completed successfully.

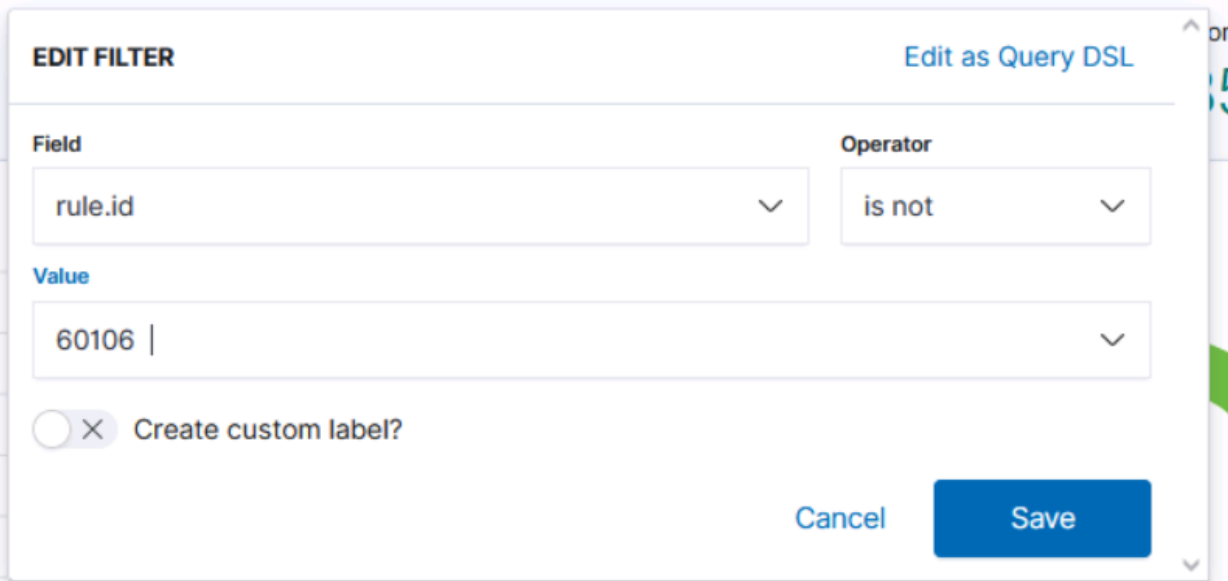
C:\Windows\system32>+++net localgroup Administrators /add Dani
'+++net' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\system32>net localgroup Administrators /add Dani
The command completed successfully.

C:\Windows\system32>
```

I typed **net users** again to see the users were deleted and **net localgroup Administrators /add Dani** which added a new user named dani.





EDIT FILTER [Edit as Query DSL](#)

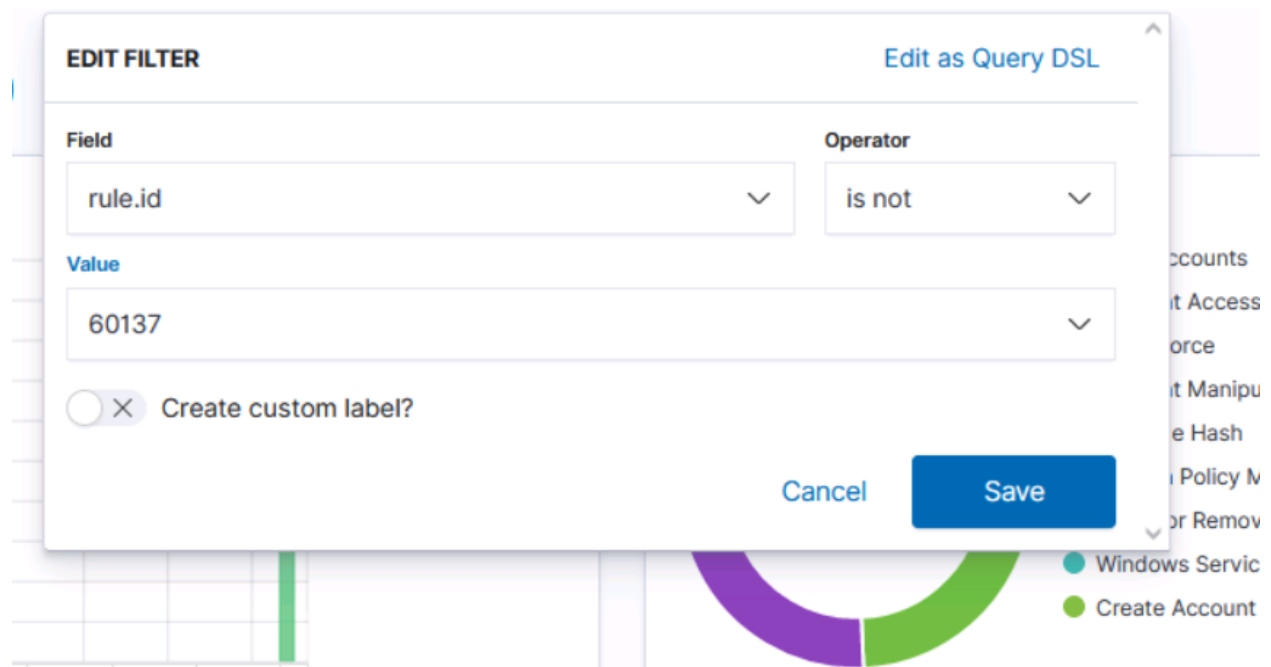
Field: rule.id Operator: is not

Value: 60106

☐ Create custom label?

Cancel Save

I then hid the security alerts for logon success and user logoff by going back to wazuh > add filter > typing **rule.id** > chose the operator as **is not** > value 60106.



EDIT FILTER [Edit as Query DSL](#)

Field: rule.id Operator: is not

Value: 60137

☐ Create custom label?

Cancel Save

Accounts
Access
Force
Manipu
Hash
Policy M
Remov

Windows Servic
Create Account

The same thing was done here but with a different value of 60137.

Security Alerts							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 25, 2025 > @ 15:51:22.583	002	DC10			Windows audit policy changed.	8	60112
Mar 25, 2025 > @ 15:51:22.549	002	DC10			Windows audit policy changed.	8	60112
Mar 25, 2025 > @ 15:51:22.533	002	DC10			Windows audit policy changed.	8	60112
Mar 25, 2025 > @ 15:51:22.520	002	DC10			Windows audit policy changed.	8	60112
Mar 25, 2025 > @ 15:51:22.520	002	DC10			Windows audit policy changed.	8	60112
Mar 25, 2025 ✓ @ 15:51:22.583	002	DC10			Windows audit policy changed.	8	60112
Table JSON Rule							
@timestamp		2025-03-25T22:51:22.583Z					
_id		GDaAz5UBdmiYobxk32IV					
agent.id		002					
agent.ip		10.1.16.1					
agent.name		DC10					
data.win.eventdata.auditPolicyChanges		Failure added					
data.win.eventdata.auditPolicyChangesId		%8451					
data.win.eventdata.category		Account Logon					
data.win.eventdata.category							

I next refreshed the wazuh page and searched for the ID 60112 and saw the numerous windows audit policy change alerts and viewed one of them more in depth.

Security Alerts 							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 25, 2025 > @ 15:52:55.744	002	DC10	T1098 T1531	Persistence, Impact	User account disabled or deleted.	8	60111
Mar 25, 2025 > @ 15:52:37.597	002	DC10	T1098 T1531	Persistence, Impact	User account disabled or deleted.	8	60111
Rows per page: 10 ▾							< 1 >

Mar 25, 2025 ▾ @ 15:52:55.744	002	DC10	T1098 T1531	Persistence, Impact	User account disabled or deleted.	8	60111
----------------------------------	-----	------	----------------	---------------------	-----------------------------------	---	-------

Table	JSON	Rule
@timestamp	2025-03-25T22:52:55.744Z	
_id	JDaCz5UBdmiYobxkTWKA	
   agent.id	002	
agent.ip	10.1.16.1	
agent.name	DC10	
data.win.eventdata.subjectDomainName	structureality	
data.win.eventdata.subjectLogonId	0xe9f31	
data.win.eventdata.subjectUserName	Administrator	

I next searched for ID 60111 and expanded the first one.

Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 25, 2025 @ 15:53:59.948	002	DC10	T1484	Defense Evasion, Privilege Escalation	Security enabled local group changed.	5	60147
TableJSONRule							
@timestamp			2025-03-25T22:53:59.948Z				
_id			KjaDz5UBdmiYobxkYmLb				
agent.id			002				
agent.ip			10.1.16.1				
agent.name			DC10				
data.win.eventdata.subjectDomainName			structureality				
data.win.eventdata.subjectLogonId			0xe9f31				
data.win.eventdata.subjectUserName			Administrator				

Security Alerts

Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 25, 2025 @ 15:53:59.933	002	DC10	T1484	Defense Evasion, Privilege Escalation	Security enabled local group member added S-1-5-21-4122946523-1810024396-762801783-1114.	5	60144

Table

JSON

Rule

@timestamp	2025-03-25T22:53:59.933Z
_id	KTaDz5UBdmiYobxkYmLb
agent.id	002
agent.ip	10.1.16.1
agent.name	DC10
data.win.eventdata.memberSid	S-1-5-21-4122946523-1810024396-762801783-1114
data.win.eventdata.subjectDomainName	structureality

I searched for IDs 60147 and 60144 and did the same thing with their expansion.

16: APPLIED LAB: Performing IoC Detection and Analysis
CySA+ (Exam CS0-003)



Congratulations, you passed!

Duration: 50 minutes, 39 seconds

Completion!